移动云 Agent Gateway 与 Agent Mesh 实践

演 讲 人

吕进

2025/11/15

KCD GROWING CLOUD NATIVE TOGETHER HANGZHOU

CHINA OpenInfra Days

CONTENT
目录

Why A New Gateway

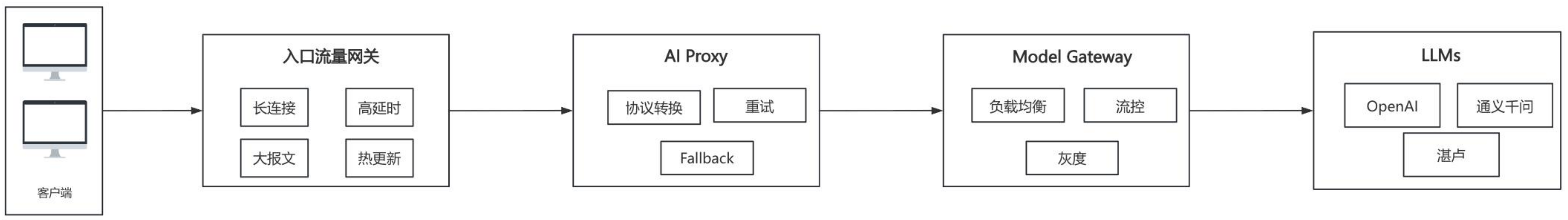
能力	API Gateway	Agent Gateway
协议	HTTP/REST、gRPC	MCP、A2A
状态管理	无状态转发	维护 agent 会话、上下文
安全控制	认证、鉴权	多租户隔离，支持工具级别授权
服务发现	无	MCP Tool、Agent CardJson
可观测性	日志/指标/链路	Token 消耗、operation

Why A New Gateway

能力	API Gateway	Agent Gateway
协议	HTTP/REST、gRPC	MCP、A2A
状态管理	无状态转发	维护 agent 会话、上下文
安全控制	认证、鉴权	多租户隔离，支持工具级别授权
服务发现	无	MCP Tool、Agent xx
可观测性	日志/指标/链路	Token 消耗、operation

为什么不基于传统网关改造？？？

LLM Proxy



客户端
入口流量网关
长连接
高延时
大报文
热更新
AI Proxy
协议转换
重试
Fallback
Model Gateway
负载均衡
流控
灰度
LLMs
OpenAI
通义千问
湛卢

CHINA
OpenInfra Days

MCP

1. Translate Stdio to HTTP SSE

```yaml
binds:
- port: 3000
  listeners:
  - routes:
    - backends:
      - mcp:
          targets:
          - name: everything
            stdio:
              cmd: npx
              args: ["@modelcontextprotocol/server-everything"]
```

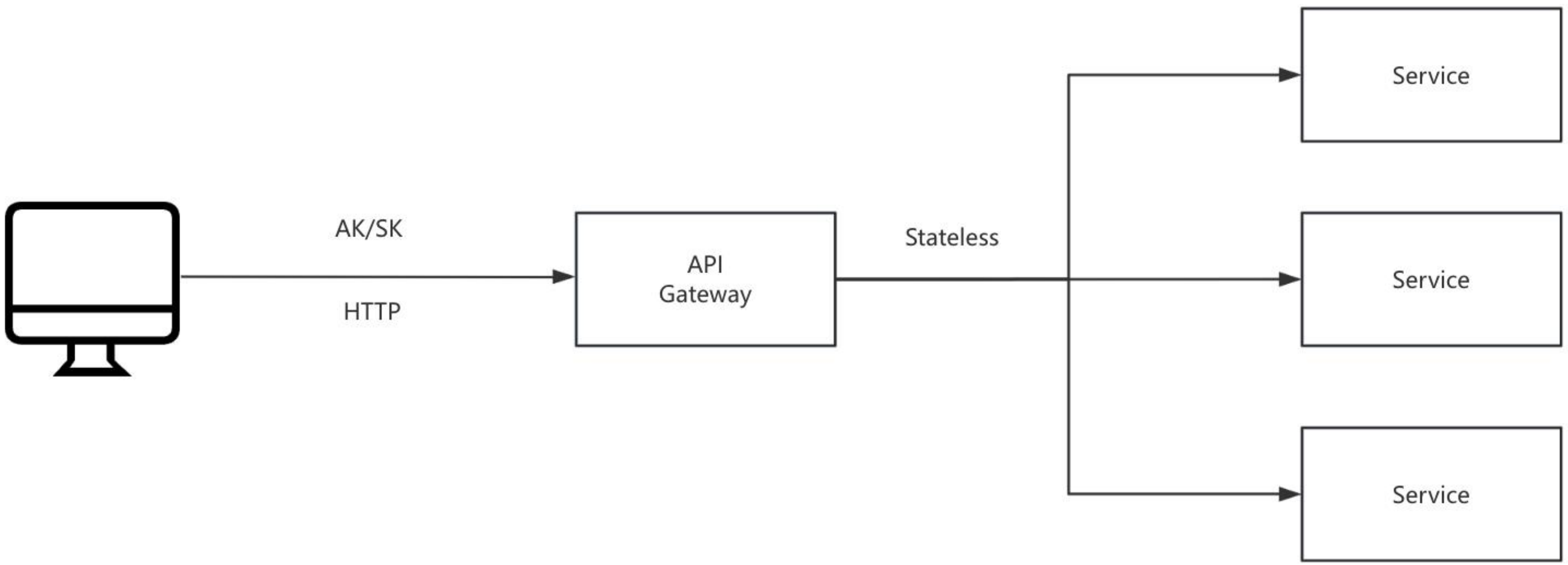
MCP

2. Translate Rest API to MCP Tools

```yaml
name: rest-amap-server
url: /amap/mcp
plugins:
  - name: apikey
    config:
      key: your-api-key-here
  - name: mcpAuthorization
    config:
      rules:
      - mcp.tool.name == "echo"
      - consumer == test && mcp.tool.name == maps-geo
tools:
  - name: maps-geo
    description: "将详细的结构化地址转换为经纬度坐标。支持对地标性名胜景区、建筑物名称解析为经纬度坐标"
    url: "https://restapi.amap.com/v3/geocode/geo"
    method: GET
    inputSchema:
      - name: address
        description: "待解析的结构化地址信息"
        type: string
        required: true
      - name: city
        description: "指定查询的城市"
        type: string
        required: false
```

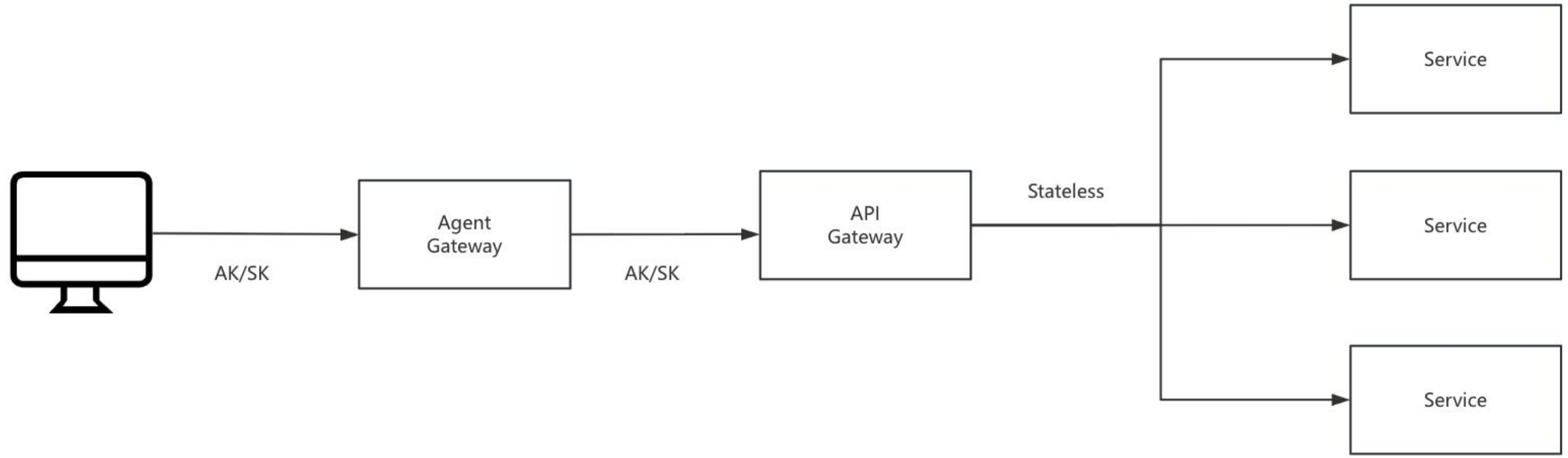
MCP

3. Translate OpenAPI(AK/SK) to MCP Tools



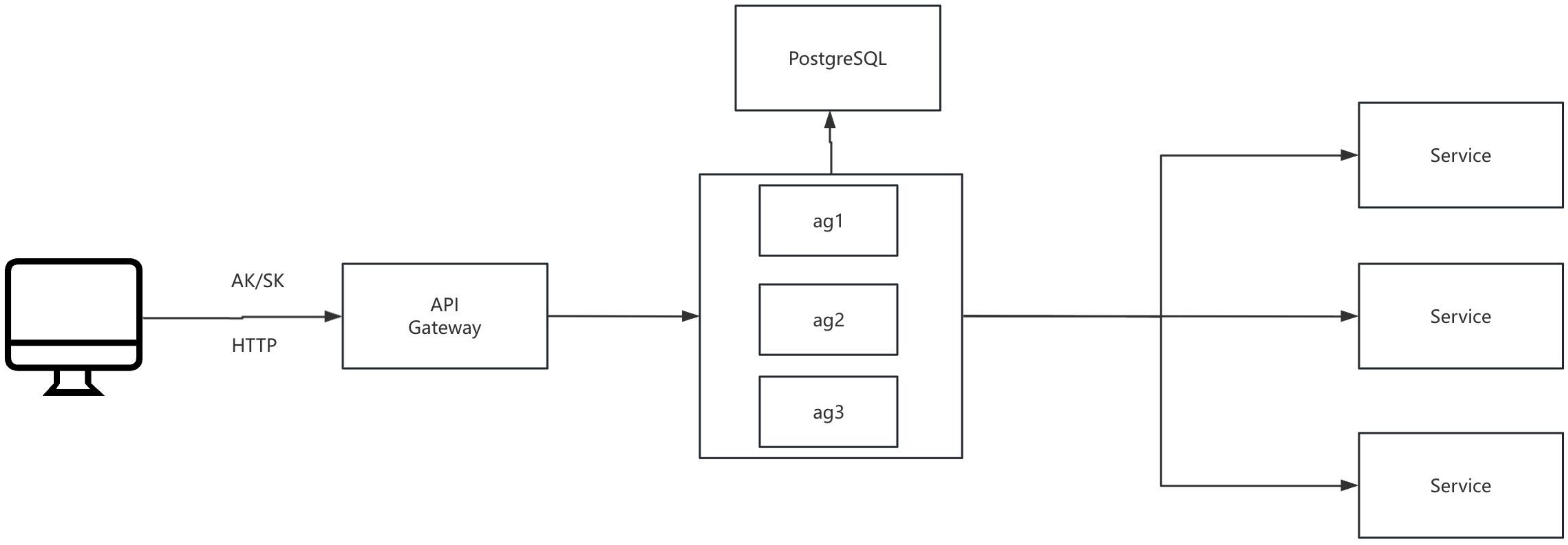
MCP

3. Translate OpenAPI(AK/SK) to MCP Tools



MCP

3. Translate OpenAPI(AK/SK) to MCP Tools



MCP

4. How to observe Stdio
   a. peekfd (linux)

```
labile@labile-hp ➜ fd $ peekfd 3538786 32

writing fd 32:
{"method":"ping","jsonrpc":"2.0","id":23}
{"method":"initialize","params":{"protocolVersion":"2025-06-18","capabilities":{},"clientInfo":{"name"
{"method":"notifications/initialized","jsonrpc":"2.0"}
{"method":"tools/call","params":{"name":"list_directory","arguments":{"path":"/home/labile"},"_meta":{'
{"method":"tools/call","params":{"name":"list_directory","arguments":{"path":"/home/labile"},"_meta":{'
{"method":"ping","jsonrpc":"2.0","id":27}
{"method":"ping","jsonrpc":"2.0","id":28}
```

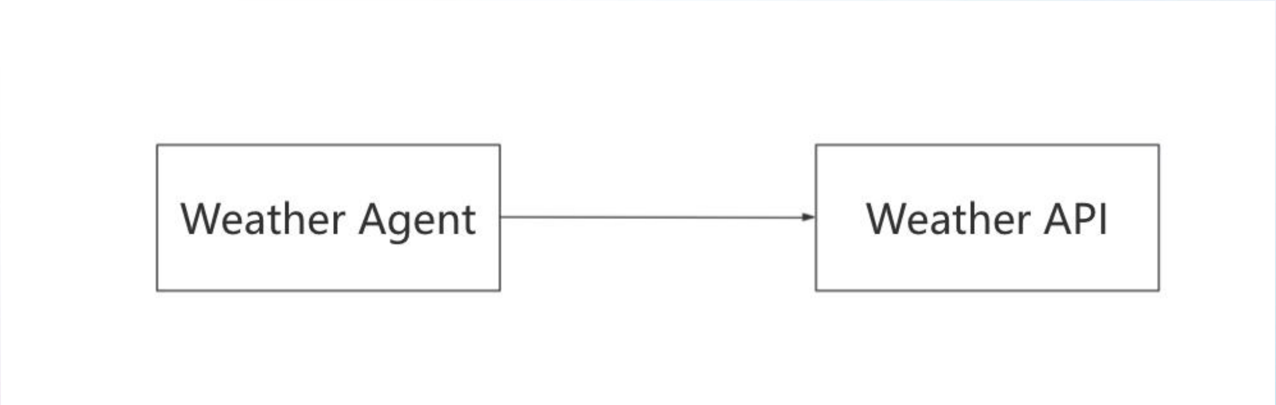
MCP

4. How to observe Stdio
    a. peekfd (linux)
    b. dtrace (macos)

```
syscall::write:entry
/ pid == $target && (int)arg0 == TFD /
{
    this->n = (int)arg2;
    this->lim = this->n < LIMIT ? this->n : LIMIT;
    printf("\n%Y write fd=%d n=%d\n%s\n",
        walltimestamp, (int)arg0, this->n,
        stringof(copyin((user_addr_t)arg1, this->lim)));
}


syscall::read:return
/ pid == $target && self->addr != 0 && (int)arg0 > 0 /
{
    this->n = (int)arg0;
    this->lim = this->n < LIMIT ? this->n : LIMIT;

    printf("\n%Y read  fd=%d n=%d\n%s\n",
        walltimestamp, TFD, this->n,
        stringof(copyin(self->addr, this->lim)));

    self->addr = 0;
}
```

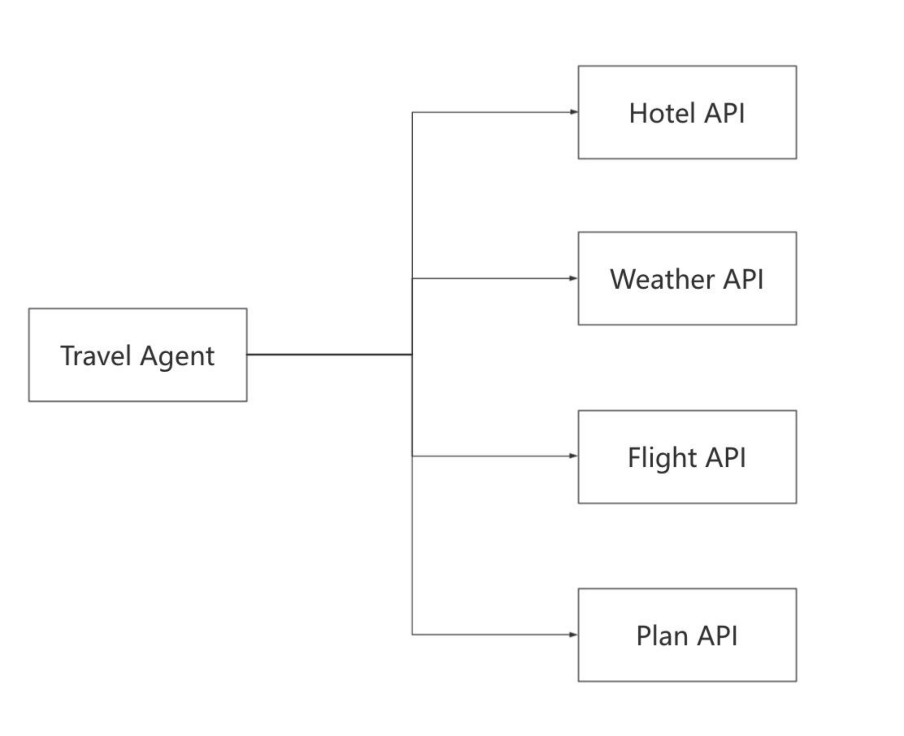
A2A

1. What is A2A ?



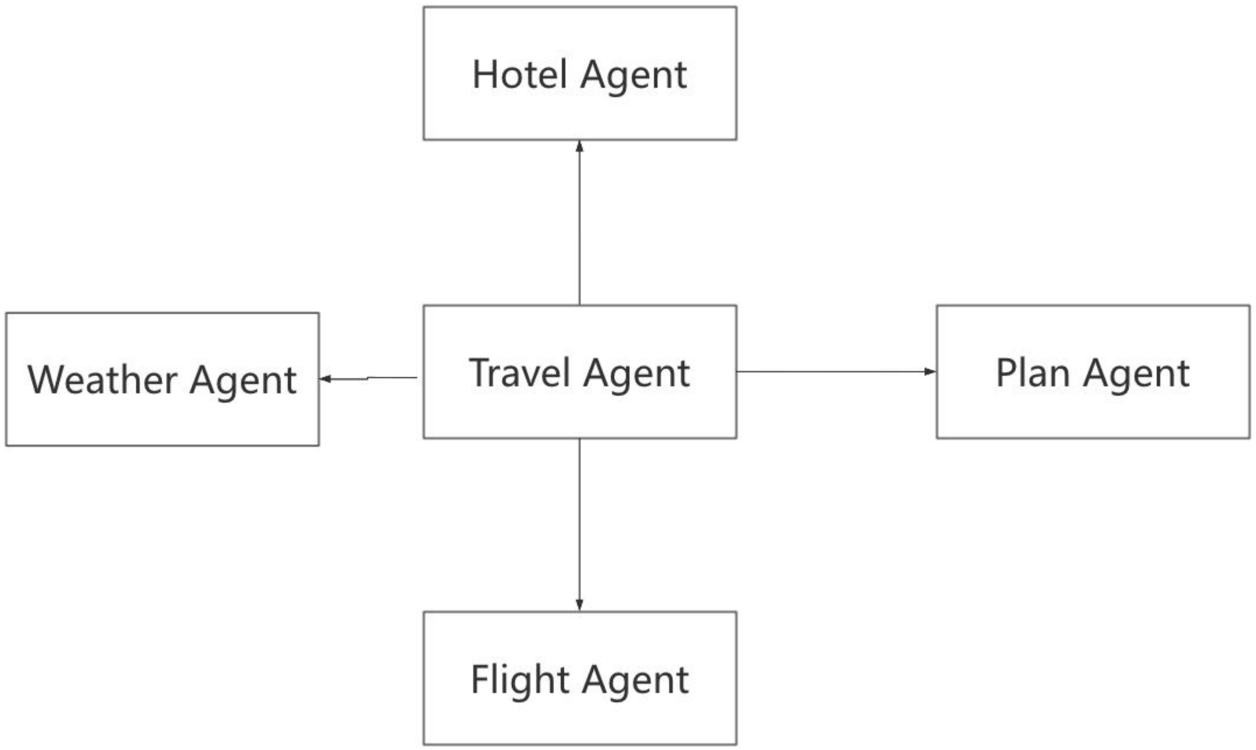
A2A

1. What is A2A ?



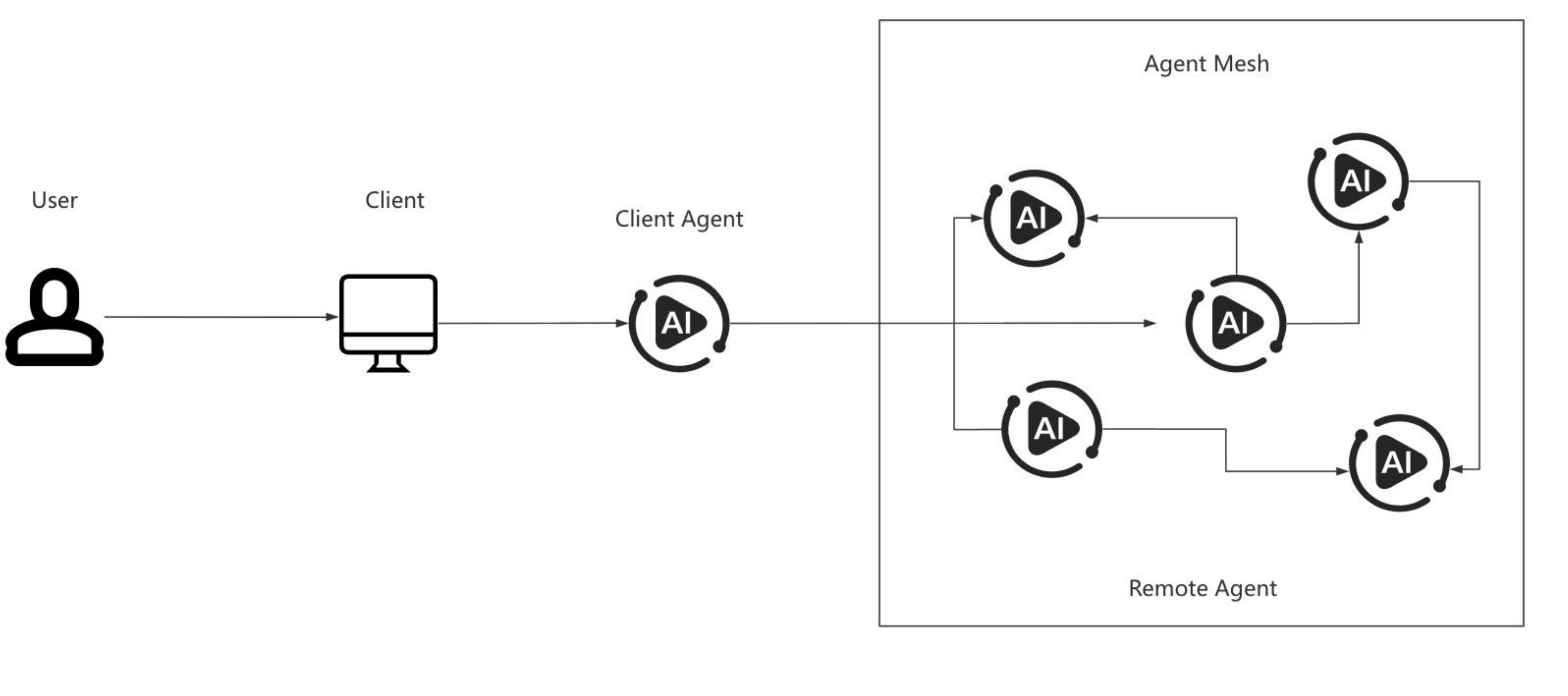
A2A

1. What is A2A ?



A2A

2. Agent2Agent(A2A) **Protocol**

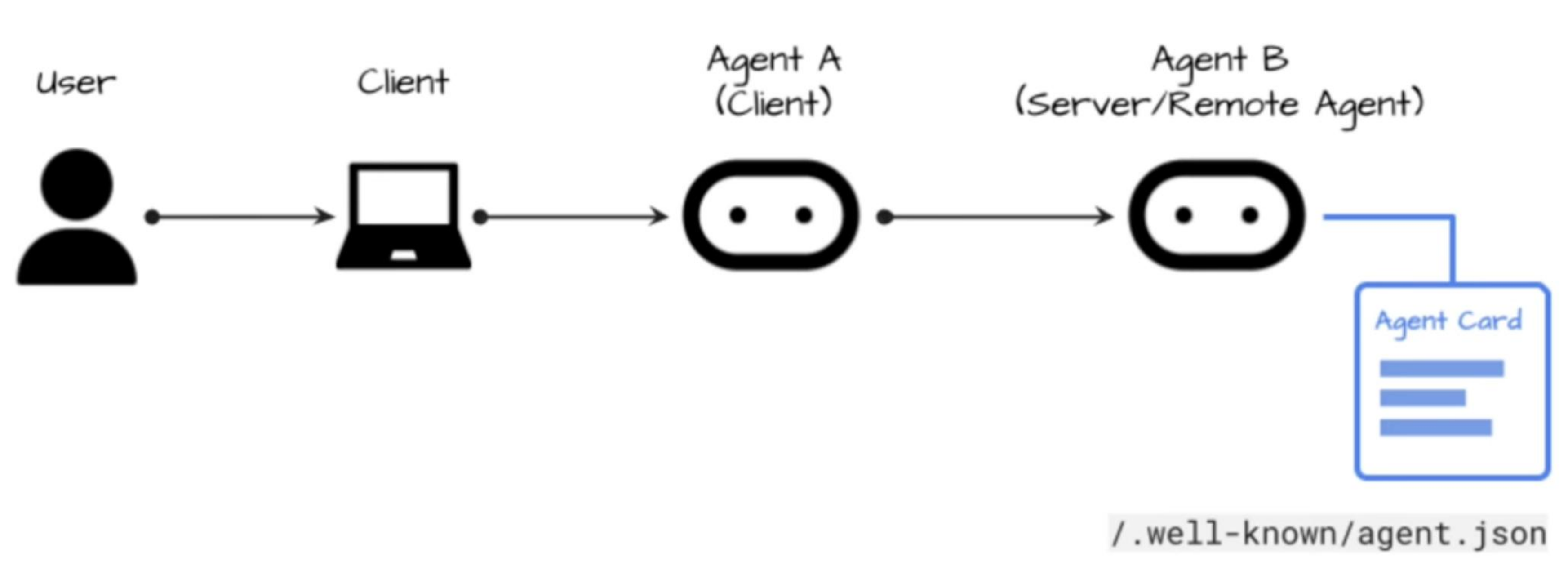


A2A

3. A2A capabilities
    a. Service Discovery
    b. Negotiation
    c. Task and State Management
    d. Collaboration

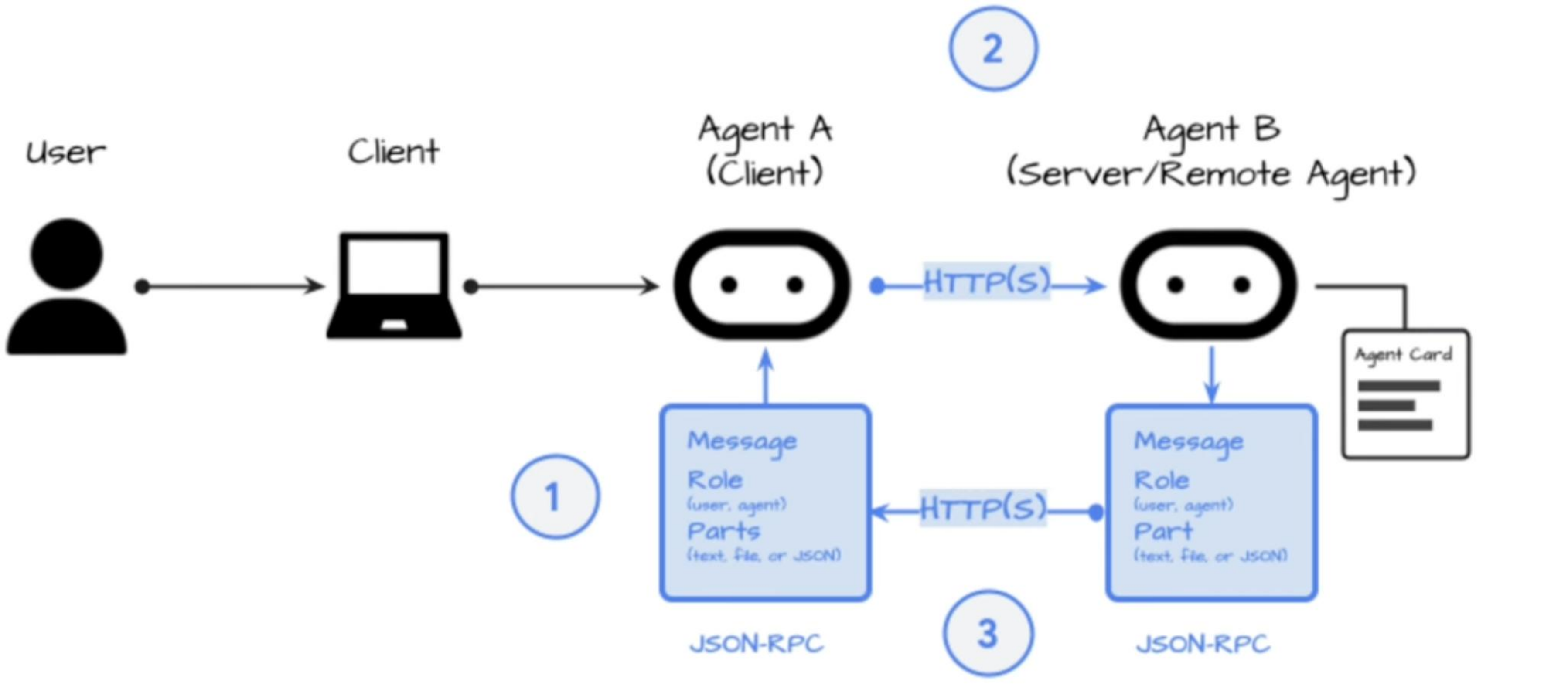
A2A

4. Agent Discover



A2A

5. Interaction



Thanks